Example 10.23

Find the area of the Standard Normal variate from – 1.96 to 0

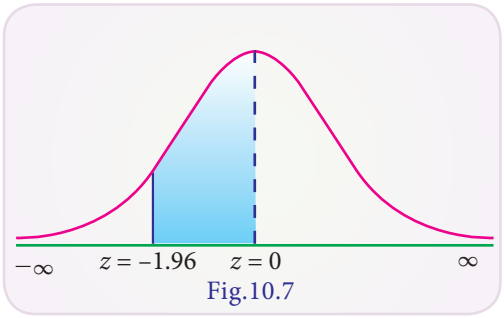


Fig.10.7

Solution:

$P(-1.96 < Z < 0) = P(0 < Z < 1.96)$ (by symmetry)

$= 0.4750$ (from the Normal Probability table)

Example 10.24

Find the area to the right of $Z = 0.25$

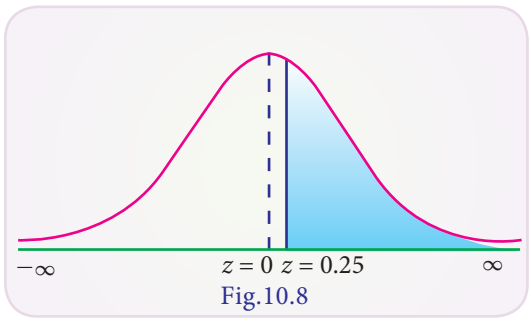


Fig.10.8

Solution:

$P(Z > 0.25) = P(0 < Z < \infty) - P(0 < Z < 0.25)$

$= 0.5000 - 0.0987$ (from table)

$= 0.4013$

Example 10.25

Find the area to the left of $Z = 1.5$

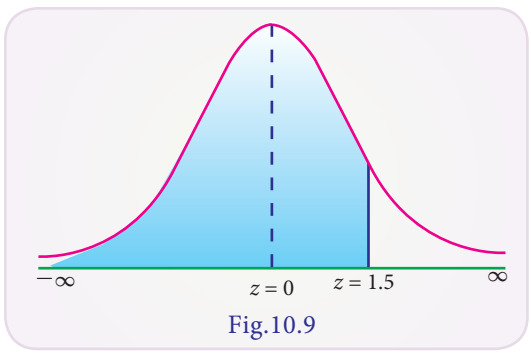


Fig.10.9

Solution:

$P(Z < 1.5) = P(-\infty < Z < 0) + P(0 < Z < 1.5)$

$= 0.5000 + 0.4332$ (from table)

$= 0.9332$

Example 10.26

Find the area between Z = -1 and $Z = 1.75$

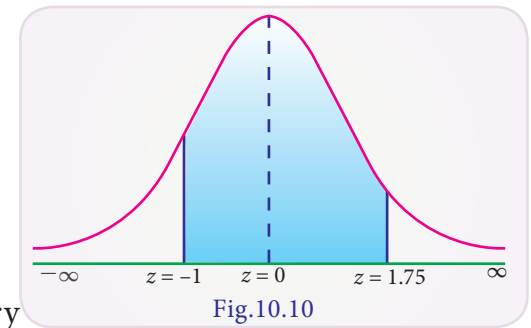


Fig.10.10

Solution:

$P(-1 < Z < 1.75)$

$= P(-1 < Z < 0) + P(0 < Z < 1.75)$

$= P(0 < Z < 1) + P(0 < Z < 1.75)$ by symmetry

$= 0.3413 + 0.4599 = 0.8012$